

Jimmy Majumder

ROBOTICS ENGINEER · AUTONOMOUS SYSTEMS · AI & COMPUTER VISION

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Summary

I am an enthusiastic and self-motivated engineering professional with transferable skills affiliated with Robotics and Mechatronics backgrounds. I have over eight years of excellent working experience attributed to my career path in research, and business development, both in corporate and voluntary settings, assigned by prestigious platforms like NASA, UNDP, the Innovation Forum, etc. I intend to work in a challenging and competitive environment where a strong sense of responsibility and commitment is required. Also, the challenges of work provide opportunities for learning and building a promising career with internal and external personnel.

Job Experience: Years 8+ (5 industry + 3 academy)

QibiTech Inc.

Tokyo, Japan

SR. ROBOTICS ENGINEER-II

Jan 2025 – Present

- **Senior Robotics Engineer Level Contributions (2026 – Present)**
- Leading development of scalable autonomous robotics platforms supporting industrial service robots and logistics automation systems.
- Architecting distributed multi-robot coordination frameworks using ROS2 communication layers and containerized services for fleet orchestration.
- Designing robot software pipelines integrating navigation, perception, and manipulation subsystems for real-world industrial deployments.
- Developing integration frameworks enabling communication between **Automated Guided Forklift (AGF) and warehouse management systems (WMS) and warehouse control systems (WCS) for nCustomers.**
- Contributing to collaborative robotics system integration initiatives with industrial partners including **Kawasaki Heavy Industries.**
- Participating in system architecture reviews and long-term robotics platform design strategies for scalable robot deployments.
- **Senior Robotics Engineer Contributions (2025)**
- Developed distributed robotics orchestration capabilities enabling coordinated operations among multiple autonomous robots operating in logistics and facility environments.
- Designed fleet-level task management pipelines for autonomous robots performing navigation, manipulation, and infrastructure interaction tasks.
- Led integration of robotics perception systems into operational robot fleets using modular ROS architectures.
- Implemented containerized robotics applications using Docker for scalable deployment across multiple robotic platforms.
- Supported robotics deployment and validation testing in industrial environments through system debugging and performance optimization, such as with our **Internal product- HATS and nCustomer's could serve.**

QibiTech Inc.

Tokyo, Japan

ROBOTICS ENGINEER

Oct 2023 – Dec 2024

- **Robotics Engineer - Mid-Senior Level Contributions (2023 – 2024)**
- Developed AI-based perception systems for robotics applications using modern deep learning architectures including YOLOv7, YOLOv8, and Faster-RCNN.
- Designed RGB-D perception pipelines integrating Intel RealSense cameras for real-time object detection and spatial awareness in robotic manipulation tasks.
- Built robotic manipulation workflows integrating perception outputs with motion planning pipelines using ROS and MoveIt.
- Worked with industrial robotic arms including Motoman manipulators developed by Yaskawa Electric for pick-and-place and service robotics applications.
- Developed robotics simulation environments using Gazebo and Unity to test robot navigation and manipulation tasks before real-world deployment.
- Contributed to robotics validation initiatives and automation demonstrations with industrial partners including **Panasonic Connect.**
- Contributed to robotics development initiatives supported by **Japanese government Org.-NEDO in collaboration with Kawasaki Heavy Industries.**
- Developed perception pipelines enabling robots to interact with building infrastructure, including elevator interfaces and environmental control systems.
- Implemented RGB-D vision systems for detecting and interacting with elevator buttons and facility devices using computer vision algorithms.
- Developed mobile manipulation systems combining autonomous navigation with robotic arm control in a warehouse of **nCustomer.**
- Built SLAM-based navigation pipelines enabling robots to autonomously move within multi-floor building environments.
- Led logistics robotics demonstrations using autonomous mobile robots for facility delivery tasks in collaboration with **Japan Post.**

QibiTech Inc.

ASSOCIATE ROBOTICS ENGINEER

Tokyo, Japan

Oct 2022 – Sep 2023

• Jr. /Asst. Robotics Engineer - Robotics Engineer- I Level Contributions (2022 – 2023)

- Joined robotics R&D team focusing on perception-driven robotic manipulation and autonomous mobile robot systems.
- Developed RGB-D perception pipelines using Intel RealSense sensors and OpenCV for robotic object detection and environment interaction.
- Implemented ROS-based control nodes integrating perception outputs with robotic manipulation and navigation systems.
- Participated in robotics research collaboration projects with researchers from **RIKEN** using DENSO Cobotta collaborative robots.
- Developed mobile manipulation prototypes combining autonomous navigation and robotic arm interaction for Humanoid AMR with **Japanese government Org.-NEDO in collaboration with Kawasaki Heavy Industries.**

Vision 2020 South Asia – UN SDG Young Leader Program

Dhaka, Bangladesh

COUNTRY DIRECTOR (OPERATIONS)

Mar 2018 – Jul 2020

- Led regional operational activities aligned with United Nations Sustainable Development Goals.
- Managed community programs involving over 5,000 members and 50,000 participants across South Asia.
- Organized social innovation campaigns and youth leadership initiatives promoting technology-driven development.

Expo Next Exhibition Pvt. Ltd.

Dhaka, Bangladesh

DEPUTY MANAGER – ENGINEERING BUSINESS DEVELOPMENT

Jul 2018 – Dec 2019

- Coordinated engineering and technology exhibitions including Bangladesh International Innovation and Development events.
- Managed international client relations and technology partner collaborations.
- Led promotion of emerging industrial and garment automation technologies.

Latest Communication Center Ltd.

Dhaka, Bangladesh

SENIOR EXECUTIVE | GROUP TEAM LEADER

Jan 2014 – Feb 2016

- Managed telecom service operations for a UK-based telecommunications provider.
- Led customer service teams and supported business development initiatives.
- Achieved operational performance targets through team coordination and process optimization.

Research Experience (3+ Years)

IASC, NASA

California, United States

PERSONAL RESEARCH ASSISTANT

Dec 2018 – Jul 2020

- Conducted research in brain-inspired robotics and space robotics systems.
- Explored human-robot interaction and cognitive control mechanisms for robotic navigation.
- Contributed to experimental studies on intelligent robotic systems for extreme environments.

SAYRID

Dhaka, Bangladesh

JUNIOR RESEARCH FELLOW

Mar 2016 – Apr 2017

- Participated in the Smart Robotics City (SRC) research initiative.
- Developed early robotics prototypes and automation concepts for smart urban environments.
- Contributed to research activities involving intelligent infrastructure and robotics integration.

Education

Kyushu Institute of Technology (Kyutech), Wakamatsu Campus

Kitakyushu, Japan

MASTER OF ENGINEERING IN ROBOTICS (GAAR PROGRAM), DEPT. OF HUMAN INTELLIGENCE SYSTEMS, LSSE

Oct. 2020 to Sept. 2022

- MAJOR PROJECTS: Assistive Robotics like Smart Robotics Walker, Clothing Assistance Robotics-HAFY
- AWARD: Government Scholar-MEXT, Excellent Thesis Presentation; Dept. Representative award from President
- GPA: 3.62 out of 4.00

World University of Bangladesh

Dhaka, Bangladesh

BACHELOR OF ENGINEERING IN MECHATRONICS, DEPT. OF MECHATRONICS ENGINEERING (MTE)

April 2014 to July 2018

- THESIS PROJECT: Smart Public Sanitation System using Automation
- AWARD: Vice Chancellor Award (semester wise)
- GPA: 3.41 out of 4.00

Research Publication: 13

CONFERENCES: 8, JOURNAL: 4, | MOST RECENT:

- **Jimmy Majumder**, John Noel Victorino, Raul Ariel Duran Jimenez, Tomohiro Shibata, “**The Study and Development of a Smart Assistive Robotic Walker System for Improved Walking Performances and Health Conditions Monitoring for Persons with Parkinson’s**”, RSJ, Tokyo, Japan, 4 pages, 4-9 Sept. 2022
- Raul Ariel D. J., **Jimmy Majumder**, Tomohiro Shibata, “**Study on cloth perception through different surfaces using active sensing in a robotic finger mechanism**”, RSJ, Tokyo, Japan, 4 pages, 4-9 Sept. 2022

- **Jimmy Majumder**, Tomohiro Shibata, “**The Study of Development of an Intelligent Smart Robotics Walker for Walking Rehabilitation and Health Monitoring Systems of PD Patients**”, International Symposium on Robotics in Industry, Agriculture and Healthcare (ISRIAH), Oral presentation, Online, 9-10 April 2022
- **Jimmy Majumder**, John Noel Victorino, Tomohiro Shibata, “**A Smart Robotics Walker System for Heart Rate Monitoring and Walking Rehabilitation of PD**”, JSME, ROBOMECH, Sapporo, Japan, 3 pages, 1-4 June 2022
- Rikuo Kawamoto, Ali Abdul, **Jimmy Majumder**, Yuichi Kurita, and Tomohiro Shibata, “**A Study on Gait Cycle Estimation Using an Integrated System of Pneumatic Artificial Muscle Assist Device and Assistive Walker for Parkinson’s Disease Patients**”, SICE SI, Kagoshima, Japan, 3F3-11, Online, 15-17 December 2021
- M. R. Islam, R. Afzal, Asaduzzaman, Ferdib-Al-Islam and **J. Majumder**, “**COV-SSDS: Design and Construction of Automatic Social Distancing System to Minimize COVID-19 Infection**,” 2022 International Conference on Advancement in Electrical and Electronic Engineering (ICAEEEE), 2022, pp. 1-6
- **Jimmy Majumder**, Arif Khan, Akramul Islam and Enamul Haq, “**Design and Construction of Smart public sanitation system**”, Advanced Materials Science Research, Global Environment and Health and Safety , International Conference on Robotics and Artificial Intelligence, Prague, Czech Republic, 20-21 July 2020
- **Jimmy Majumder**, etal., “**Robot Navigation For Blind People Using Multi Sensor**”, International Journal of Scientific Engineering Research (IJSER), 2018, ID: I0100691
- **Jimmy Majumder**, etal., “**Stick Robot for Blind People Using Multi-Sensors**”, International Journal of Advance Research (IJAR), 2nd Jun 2018, ID: IJAR-23487
- **Jimmy Majumder**, “**Navigation For Blind People using Stick Robot**”, Mechatronics Engineering, World University of Bangladesh, 1st edition, April 2017
- **M. Jimmy**, Fedouse. F., Majumder. MD.Rashed, “**Stick Robot for Blind People Using Multi-Sensors**”, 204th International Conference on Science, Engineering and Technology-ICSET, Kyoto, Japan, 3 pages, 09-06-2018
- **Jimmy Majumder**, etal., “**Autonomous AI Based Robot Navigation for Smart Fish Catching Process Based on Embalmed System**”, IEEE-ICSET, Dhaka, Bangladesh, 3Pages, December 22–24, 2017

Robotics Systems Development (50+ Projects)

• Industrial Autonomous Robotics Platforms

- Autonomous Mobile Robots (AMR) & Automated Guided Forklift (AGF) for warehouse logistics
- Mobile manipulation systems integrating robotic arms with AMR navigation
- Humanoid upper-body AMR for elevator door operation (NEDO robotics project)
- Infrastructure interaction systems for doors, buttons, and building facilities
- Industrial teleoperation systems for factory machinery and engine operations
- Pick-and-place robotic manipulation pipelines for logistics automation
- Warehouse cart delivery robots and automated material transport systems
- Fleet-level coordination platforms for multi-robot industrial deployments

• Assistive Robotics and Healthcare Systems

- Smart Robotic Walker for mobility assistance and rehabilitation
- World’s first dressing assistive robot - C(HAFY) for dressing support
- Assistive hover using HEBI Robotics Actuators - R & D
- Health monitoring and rehabilitation robotics systems
- Using HATS Highly Autonomous Teleoperation System, Mobile robot operations in a Hospital
- Smart sanitation and hygiene robotics platforms

• AI-Driven Autonomous Robotics

- Humanoid robot manipulation and autonomous navigation
- Brain-signal-controlled robot navigation using BCI
- AI-based robotic perception systems using NVIDIA Jetson platforms
- Autonomous driving robots using TurtleBot3
- AI-enabled food detection and service robotics applications
- Intelligent home security robots with vision-based detection

• Agricultural and Environmental Robotics

- Automated harvesting robotic systems
- Smart fish feeding automation platforms
- AI fish detection and hunting robots
- Solar tracking systems for renewable energy optimization

• Early Innovation and Educational Robotics

- Multi-sensor blind navigation robot
- Firefighter robot for hazard detection
- Defense security robotic surveillance systems
- Bangla-speaking humanoid robot prototype
- Home automation robotics systems
- IPD-based autonomous line follower robots

Technical Skills

Industrial Robots Used	Baxter, HEBI ARM, UR3, Toyota's HSR, Denso Cobotta, KHI's Nyokkey Gen 2 & 3, S-CART, Motoman, Kachaka etc. (20+)
Hands-On Components	Industrial Actuators, Multi-Sensors, RealSense, 3D Vision Camera, Kinect V2, Arduino, Jetson, Raspberry Pi
APIs & Libraries	ROS 1/2, MoveIt, OpenCV, Keras, NumPy, PyTorch, Pandas, TensorFlow, Hugging Face, Microsoft Virtual Camera API
Visualization	Gazebo, Rviz, Unity, Jupyter Notebook, VSLAM, GSLAM, PlantUML, Doxygen, JaamSim, SimPy
Tools & IDE	MATLAB, Visual Studio, LaTeX, Proteus, WinAVR, AutoCAD, Vim, Microsoft Azure, Docker, Doxygen
AI & ML	NVIDIA CUDA, YOLO, COCO, CNN, R-CNN, Fine-Tuning, Hugging Face, RL
Programming	Python, C++, Embedded C / C, Java

Extracurricular Activities

Present- Venture & Innovation Strategist , Google for Startups	<i>Tokyo, Japan</i>
Present- Founder & President , Bangladesh Advance Robotics Research Center (BARRC)	<i>Dhaka, Bangladesh</i>
Present- Member , Robotics Society of Japan (RSJ)	<i>Japan</i>
Present- Member , Japan Society of Mechanical Engineers (JSME)	<i>Japan</i>
Present- Editorial Board Member , Asia Pacific Journal of Advanced Education and Technology (APJAET)	<i>Philippines</i>
2020- Board of Advisors , Waste Management Project of ABC Construction Chemical Company Ltd..	<i>Bangladesh</i>
2022-2025 Mentor , Japan-Bangladesh Startup Projects, ICT Division	<i>Bangladesh / Japan</i>
2021-2022 Vice-President , Foreign Students Association, Kyushu Institute of Technology	<i>Kitakyushu, Japan</i>
2020-2022 Executive Board Member , Innovation Forum – Bangladesh Branch	<i>London, UK</i>
2019-2020 Director, Organizing Committee , The Robotics Society of Bangladesh	<i>Dhaka, Bangladesh</i>
2017-2020 SDG Young Leader Program Participant , UNDP Vision 2020 South Asia	<i>South Asia</i>
2016-2018 Convener & Vice-President , WUB Mechatronics Club	<i>Dhaka, Bangladesh</i>
2016-2017 Joint Secretary , WUB Photography & Media Club	<i>Dhaka, Bangladesh</i>
2015-2018 Advisor – Advance Robotics , Bangladesh Science Society	<i>Dhaka, Bangladesh</i>

Honors & Awards

25+ AWARDS | LIVE SESSIONS: 30+ | STUDENTS ENGAGED: 10,000+ | ONLINE PARTICIPANTS: 1M+ | ARTICLES & BLOGS: 40+

KEY ACHIEVEMENTS & NOTABLE RECOGNITIONS

2020	MEXT Scholarship (Monbukagakusho) , Japanese Government Graduate Scholarship for Robotics Research at Kyushu Institute of Technology	Japan
2022	Excellent Thesis Presentation Award , Graduate School of Life Science and Systems Engineering, Kyushu Institute of Technology	Japan
2022	Graduate School Special Recognition , Selected Graduate Representative – Kyushu Institute of Technology	Japan
2021	Jetson AI Specialist , NVIDIA Deep Learning Institute – AI Certification for Jetson Embedded AI Development	Kyushu Institute of Technology, Japan
2018	Best Thesis Award – B.Sc. Mechatronics Engineering , World University of Bangladesh	Bangladesh
2016	Best Performer Award , Vice Chancellor Award – World University of Bangladesh	Bangladesh
2019	Global Business Excellence Award , International Business Leadership Recognition	Global
2022	Space Excellence Award , International Space Research Recognition	Global

Selected Media & Demonstrations

2025	Autonomous Teleoperation for Industrial Robots — HATS Development Interview  , Jimmy's Interview as a Robotics Engineer at QibiTech	Tokyo, Japan
2025	Automated Food Delivery Demonstration with Mobile Robots and Kawasaki Heavy Industries at KAWARUBA  , PR TIMES — QibiTech Press Release	Tokyo, Japan
2025	HATS Platform Adds Remote Recovery Functionality for Manipulators  , PR TIMES — QibiTech Press Release	Tokyo, Japan
2025	Robotics Technology Integration Platform HATS Updated with Fleet Management Capability  , PR TIMES — QibiTech Press Release	Tokyo, Japan
2025	Lighting the Path: How Two WUB Alumni Are Shaping Global Industries  , The Daily Star	Bangladesh
2023	Autonomous Mobile Robots with Arms for Automated Delivery — Remote Control Service HATS Research (NEDO Project)  , PR TIMES — QibiTech Press Release	Tokyo, Japan
2023	Humanoid Upper-Body AMR Demonstration for Elevator Door Operation (NEDO Project)  , QibiTech Inc. – Robotics Demonstration with Kawasaki Heavy Industries	Tokyo, Japan
2023	Autonomous Restaurant Delivery Robot Door Operation Demonstration  , QibiTech Inc. – NEDO Smart Delivery Robotics Project	Tokyo, Japan
2022	Assistive Dressing Robot (HAFY) Demonstration — IREX Robotics Exhibition  , Shibata Laboratory, Kyushu Institute of Technology	Tokyo, Japan
2025	Preparing for a Robotics Career in Japan — Interview with Jimmy Majumder  , Ajker Patrika — Interview in Bangla Language	Bangladesh
2018	Dreaming of a Robot-Driven Digital Bangladesh — Feature on Jimmy Majumder  , Bangladesh Pratidin — Feature Story (Bangla)	Bangladesh
2018	Jimmy Majumder: A Bangladeshi Young Star in the Firmament of Tech  , The Daily Observer	Bangladesh

References

AVAILABLE UPON REQUEST